Paleomagnetism and Age Determination of Cretaceous Rocks From Gyeongsang Basin, Korean Peninsula

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Paleomagnetic measurements and age determinations have been made on samples of sediments and lavas of Cretaceous age from the Gyeongsang Basin, South Korea. The K-Ar age of 68.1 ± 3.4 m.y. for basalts from the Hagbong Volcanic Member confirms the Cretaceous period of Gyeongsang Supergroup. Paleomagnetic directions are fairly grouped with a mean confidence interval of declination of 26.6°, inclination of 62.3°, and α_{95} = 8.3°. The reliability of the direction is ascertained through thermal and alternating field demagnetization techniques and comparison of the remanent directions of sedimentary rocks and a lava flow. The South Korean pole obtained (191°E, 69°N) is in excellent agreement with the poles from neighboring regions of North Korea, China, Kolyma, and the Siberian Platform. From this concordance it is inferred that since the Late Cretaceous the Korean Peninsula has been a part of the Asian continent and was not subjected to any tectonic rotation relative to the Asian continent. This high stability of the Korean Peninsula against tectonic disturbances is explained by the presence of a thick and stiff lid within the lithosphere, inferred from the comparison between the character of the lithosphere beneath the Korean Peninsula and Japan.

Introduction

Paleomagnetism, marine magnetic anomaly, and seismological studies document large-scale relative displacements or rotations of peninsulas to their host continents. The Indian Peninsula has been colliding with the Asian continent since the Paleogene [*Molnar and Tapponnier,* 1975, 1977], and the Arabian Peninsula is now colliding at the Zagros region of Iran [*Bird et al.,* 1975]. The Iberian, Italian, and Malay peninsulas have all been subjected to tectonic rotations [*Van der Voo,* 1969; *Lowrie and Alvarez,* 1974; *McElhinny et al.,* 1974]. Since peninsulas are situated at the edge of a continent, where crust is thin [*Cummings and Shiller,* 1971], the interactions among plates may easily cause the relative motion of the peninsula with respect to the host continent.

The Korean Peninsula is one of the largest peninsulas in the eastern margin of the Asian continent. The work presented here is an attempt to obtain the relative movement between the Korean Peninsula and the Asian continent from the viewpoint of paleomagnetism.

The Cretaceous strata of Gyeongsang Basin in South Korea are very suitable for paleomagnetic study, because the strata consist of many kinds of rocks (lava, tuff, shale, and sandstone) with different kinds of natural remanent magnetization (NRM) and their bedding planes of strata are clearly recognized. The present paleomagnetic work could therefore provide more reliable paleomagnetic directions than the previous works from igneous rocks [*Kienzle and Scharon,* 1966; *Ito and Tokieda,* 1980].

Geology and Sampling

Gyeongsang Basin is located in the southeastern part of the Korean Peninsula (Figure 1). This basin consists of a thick sequence of fluviolacustrine sediments intercalated with lavas and volcanoclastic rocks of Cretaceous age. These strata constitute the Gyeongsang Supergroup. According to the definition of *Chang* [1975] the Gyeongsang Supergroup is divided into three groups: the Sindong, Hayang, and Yuchon groups in ascending order (see Table 1). The Sindong Group consists of sediments of the prevolcanic phase, the Hayang Group comprises nonvolcanic sediments with some volcanic horizons, and the Yuchon Group consists of volcanic formations.

The Sindong Group, the lowest of the Gyeongsang Supergroup, is extensively fossiliferous, and it is documented as lowest Cretaceous in age [*Yabe,* 1905; *Kawasaki,* 1928; *Um and Reedman,* 1975]. The K-Ar data for granitic plutons intruding the Gyeongsang Supergroup range from 62 to 88 m.y. [*Chang,* 1975; *Lee,* 1980], which puts the end of the Cretaceous as the upper limit of the supergroup. These data indicate that the Gyeongsang Supergroup was formed during the Cretaceous.

More than 120 samples were obtained from 10 widely

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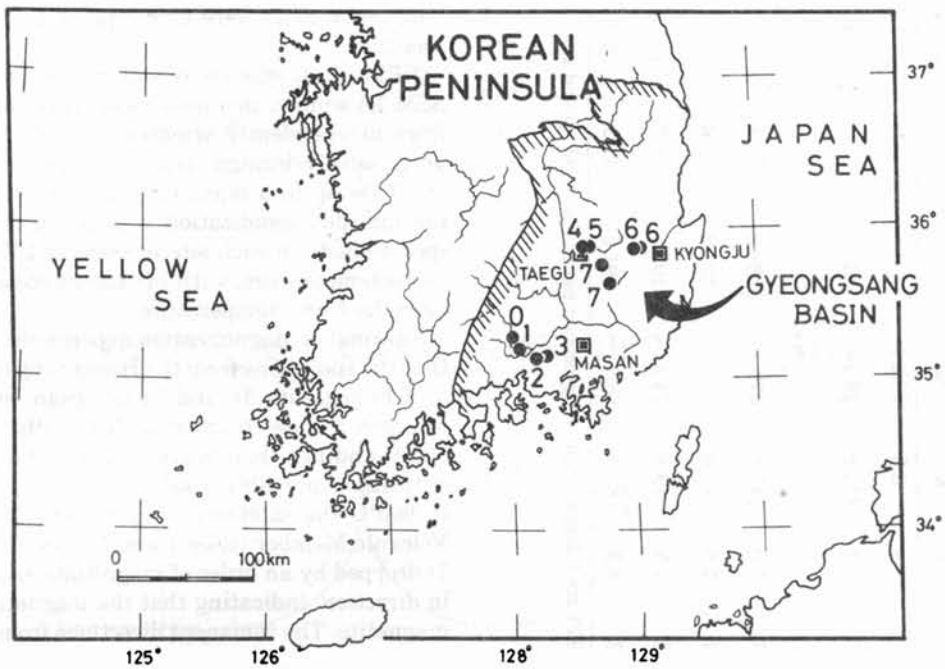


Fig. 1. Sampling localities in Gyeongsang Basin: site 0, Nagdong Formation; site 1, Hasandong Formation; site 2, Jinju Formation; site 3, Chilgog Formation; sites 4 and 5, Hagbong volcanic members; site 6, Chaeyagsan Formation; and site 7, Yuchon Group. Stratigraphic horizons of sampling sites are given in Table 1.

spaced locations in the Gyeongsang Basin. Figure 1 shows the distribution of sampling sites. The samples at sites 0–2 are from the Sindong Group, those at sites 3–6 are from the Hayang Group, and those at site 7 are from the Yuchon Group. The bedding planes were clearly recognizable at all sampling localities except for site 5 (of basaltic lavas). Detailed stratigraphic horizons of sampling sites and the rock types are given in Tables 1 and 2. The basaltic rock samples were collected for K-Ar age determination from the Hagbong Volcanic Member of site 4, in the Hayang Group.

Paleomagnetism

The block samples collected were drilled into 2.4-cm-diameter cores and cut into individual specimens 2.4 cm

TABLE 1. Stratigraphy of Gyeongsang Supergroup in Gyeongsang Basin and the Stratigraphic Horizons of Sampling Sites

Group	Subdivision	Site
Yuchon Group		7
Hayang Group		
Jindong Subgroup	Geoncheonri Formation	
	Chaeyagsan Formation	6
	Songnaedong Formation	
	Banyaweol Formation	
Uiryong Subgroup	Haman Formation	
	Hagbong Volcanic Member	5 and 4
	Silla Conglomerate	
Chilgog Subgroup	Chilgog Formation	3
Sindong Group	Jinju Formation	2
	Hasandong Formation	1
	Nagdong Formation	0

Data after *Chang* [1975].

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TABLE 2. Results of Paleomagnetic Measurement

	Division	Locality		Rock Type	N	Level of Demagnetization	D, deg	I, deg	α_{95}, deg	K	Virtual Geomagnetic Pole (North)		∂p, deg	∂m, deg
		Latitude, °N	Longitude, °E								Latitude, °N	Longitude, °E		
Site 7	Yuchon Group	35.72	128.74	andesite	22	200 Oe	20.8	57.7	19.4	3.5	73.2	203.6	20.9	28.5
Site 4	Hagbong Volcanic Member	35.89	129.03	basalt	6	300 Oe	6.6	66.3	5.1	142.8	76.3	147.3	8.4	8.4
Site 3	Chilgog Formation	35.17	128.16	shale	27	200 Oe	27.8	58.7	1.9	210.8	67.5	198.8	2.1	2.8
Site 2	Jinju Formation	35.16	128.10	sandstone	7 24	150°C 300 Oe	53.1	60.1	7.5	12.9	48.4	193.2	8.6	11.4
Site 1	Hasandong Formation	35.17	128.03	sandstone	20	300 Oe	18.8	58.8	7.5	19.8	74.5	196.1	8.3	11.2
Mean		37.	128.		5		26.6	62.3	8.3	85.0	68.9	191.2	10.1	12.9

Parameters are as follows: N, number of specimens; D, declination; I, inclination; α_{95}, radius of 95% confidence circle; K, precision parameter; and ∂p and ∂m, semiaxes of ovals of 95% confidence.

long; most cores were long enough to yield two or three specimens.

NRM of all specimens was measured on a Schonstedt SSM-1A spinner magnetometer. Three or more specimens from independently oriented samples in each site were progressively demagnetized in alternating field (af) in steps of 50 Oe up to a maximum field of 600 Oe. Progressive thermal demagnetization was also undertaken on three specimens from each site in steps of 100° or 150°C to the Curie temperature, with increment steps reduced to 50°C near the Curie temperature.

Thermal demagnetization experiments (Figure 2) show that the specimens from the Hasandong (site 1), Jinju (site 2), Chilgog (site 3), and Chaeyagsan (site 6) formations retained a large fraction of NRM after being heated to 650°C and lost their magnetization at 700°C. Their magnetizations probably reside in the hematite. After heating to 600°C the intensity of specimens from the Hagbong Volcanic Member (sites 4 and 5) and Yuchon Group (site 7) dropped by an order of magnitude and became random in direction, indicating that the magnetization resides in magnetite. The remanent directions from all sites differed only slightly from the NRM direction before treatment during thermal cleaning up to near the Curie temperature. Thermal cleaning experiments demonstrate that the specimens have a single and stable component of magnetization.

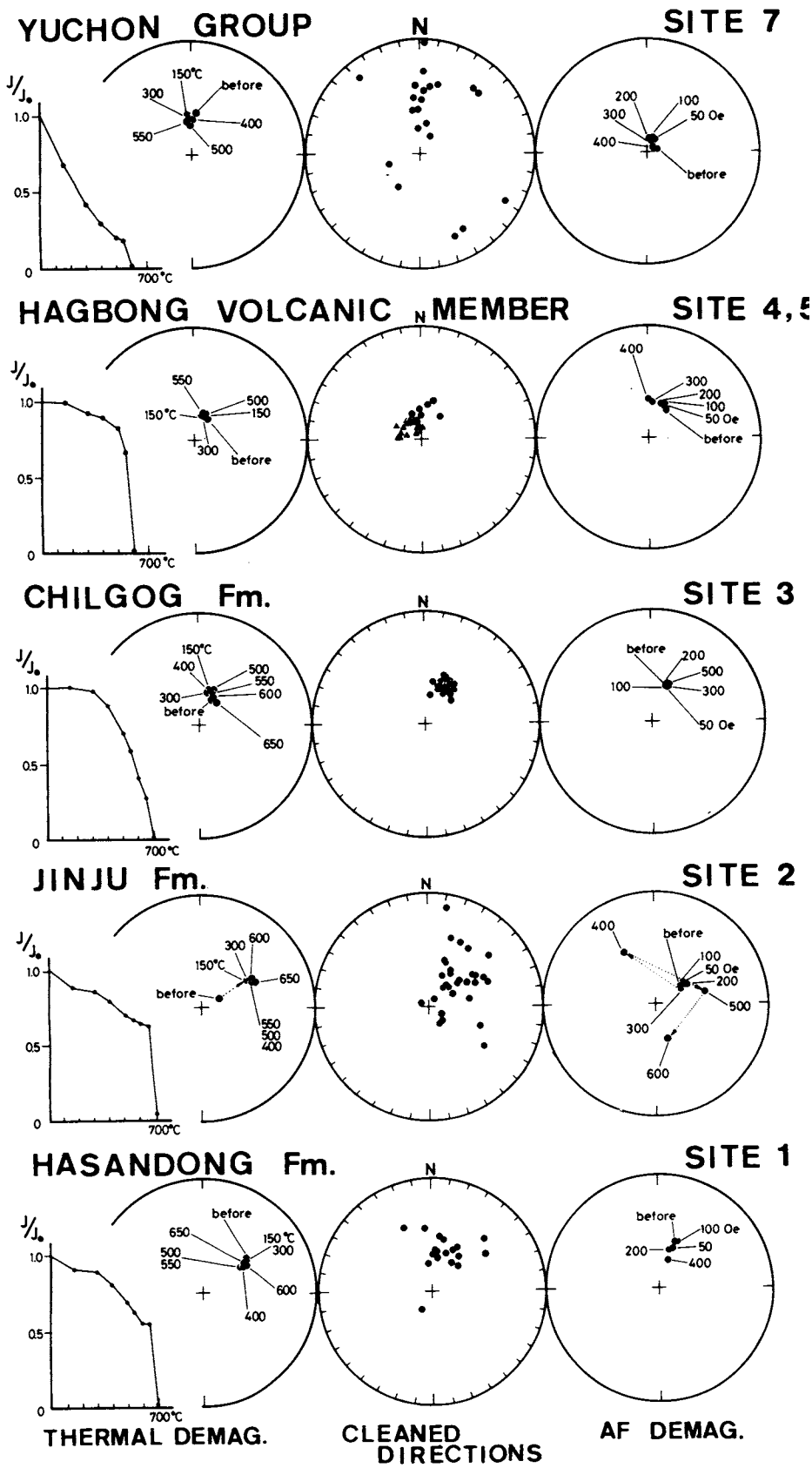
There is little difference in the direction changes after alternating field and thermal demagnetization for pilot specimens from the Hasandong Formation, Chilgok Formation, Hagbong Volcanic Member, and Yuchon Group (Figure 2). The optimum demagnetization was confined to the af demagnetization method; the optimum demagnetization level for each site was chosen as the value for which more than three directions of pilots were most clustered. The NRM's after cleaning by optimum demagnetization fields are drawn in Figure 2.

Remanent magnetizations of specimens from the Jinju Formation (site 2) showed a different behavior in response to af and thermal demagnetization. Specimens were cleaned either by af demagnetization in a peak field of 300 Oe or by thermal demagnetization at 150°C. The site mean was calculated from both af and thermal demagnetization data, because there was little difference in direction between the two data sets.

Specimens from the Nagdong Formation (site 0) had a remanent intensity of less than 10^{-8} emu/g. These specimens were rejected to avoid the introduction of errors through instrumental noise on measurements. Pilot specimens from the Chaeyagsan Formation (site 6) showed a stable remanent magnetization in response to both af and thermal demagnetization. However, the precision parameter k [*Fisher*, 1953] of three pilot specimens was less than 2.0 at any demagnetization level. The directions of all specimens (N = 12) demagnetized after 300 Oe were also highly scattered with a precision parameter of only 1.3. It

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TABLE 3. Results of K-Ar Dating for Volcanics From Hagbong Volcanic Members

Sample	Locality	Isotope Age, m.y.	Scc $^{40}Ar/g \times 10^{-5}$	^{40}Ar, percent	K, percent
PG 35	129.03°E, 35.90°N	68.1 ± 3.4	0.295	74.8	1.10
			0.298	69.8	1.10

The constants for the age calculation are $\lambda_\beta = 4.962 \times 10^{-10}$ yr^{-1}, $\lambda_\epsilon = 0.581 \times 10^{-10}$ yr^{-1}, and $^{40}K = 1.167 \times 10^{-4}$ atom per atom of natural potassium [*Steiger and Jager,* 1977]. The error indicated for the age consists of a summation of all analytic errors (5%).

is possible that the block samples from this site may have been subjected individually to arbitrary rotation or movement after they had acquired the stable remanent magnetization. Data from this site were therefore omitted from further paleomagnetic considerations.

Table 2 represents a summary of the paleomagnetic results after correction for geological dip. The largest dip correction was 15°; most were in the range of 11°. The characteristic NRM direction of specimens from site 5 of the Hagbong Volcanic Member ($D = -19.0$, $I = 789.9$, and $\alpha_{95} = 6.9$) deviates by 22.2° from the mean of the Gyeongsang Supergroup, while the deviation of the NRM direction of site 4, from the same volcanic member, is only 9.9° from the mean. The large deviation of site 5 is ascribed to the lack of the dip correction for direction. This result documents the importance of the dip correction for obtaining reliable paleomagnetic directions.

The mean declination and inclination values were calculated from five directions into which the directions at five sites in the Gyeongsang Basin were converted at the representative point of the Korean Peninsula (128°E, 37°N). The Gyeongsang Supergroup yields a mean direction of $D = 26.6°$, $I = 62.3°$, and $\alpha_{95} = 8.3°$ at the representative point. The reliability of the direction is ascertained through thermal demagnetization, dip correction, and the agreement of directions between lava flow and sediments, although no specimens with reversed polarity were observed.

K-Ar Age of Basalts From the Hagbong Member

K-Ar whole rock age of basalt from the Hagbong Volcanic Member in the Hayang Group was determined by the Geochronometry section of Teledyne Isotope. Prior to the request for K-Ar dating, thin sections of the samples were examined. The rock is composed of phenocrysts of olivine and clinopyroxene and groundmass of clinopyroxene, plagioclase, opaque mineral, biotite, and less than 3% glass. Although all olivine crystals and some glass are altered to talc (sometimes iddingsite) and chlorite, respectively, the other constituents of rock, particularly interstitial phases, remain unaltered. According to criteria discussed by *Mankinen and Dalrymple* [1972] this rock is relatively suitable for dating.

Results of K-Ar dating are given in Table 3. The age obtained, 68.1 ± 3.4 m.y., is consistent with the geological estimation [*Chang,* 1975] that the Gyeongsang Supergroup was formed during the Cretaceous.

Significance of Results

The most striking result from this work is the great predominance of normally magnetized rocks. The question arises whether this predominance of normal polarity is primary or due to overprinting by the later heat of the intrusion of the plutons known as the Bulgugsa Granite Series [*Um and Reedman,* 1975]. The thermal demagnetization tests strongly suggest that no significant thermal overprinting has occurred and the magnetization is primary. The radiometric data support this implication: Since the age determination shows that the Gyeongsang Supergroup was formed during the Cretaceous, rocks could acquire magnetization with normal polarity, which was dominant in the middle to later Cretaceous age [*Larson and Hilde,* 1975; *Labrecque et al.,* 1977; *Lowrie and Alvarez,* 1981].

No systematic movements are observed in the distribution of virtual geomagnetic pole positions for five acceptable sites as shown in Figure 3. Radiometric data and the stratigraphic sequence suggest that the rock units sampled covered nearly the entire range of Cretaceous time. It seems reasonable therefore to assume that the mean pole position obtained (191°E, 69°N) averages over enough time to eliminate secular variation effects and gives a reliable estimate of the Cretaceous paleomagnetic pole position.

Comparison of the pole position with other Cretaceous pole positions from the Korean Peninsula is made in Figure 3. Overlapping circles of 95% confidence indicate that the Gyeongsang Supergroup pole and other poles from volcanics [*Kienzle and Scharon,* 1966], granites [*Ito and Tokieda,* 1980], and North Korean sediments [*Gurarii et al.,* 1966] are not significantly different and probably repre-

Fig. 2. (Opposite) Equal-area projections showing cleaned directions (center) of remanent magnetization together with examples of thermal (left) and alternating field (right) demagnetization of specimens. Solid symbols are lower hemisphere plots. Optimum temperature and field strength of the demagnetization are listed in Table 2. The remanent directions without tilting correction in Hagbong volcanic members are shown as triangles. The decay of remanent magnetization intensity is also plotted versus temperature during thermal demagnetization: J_o is the initial intensity.

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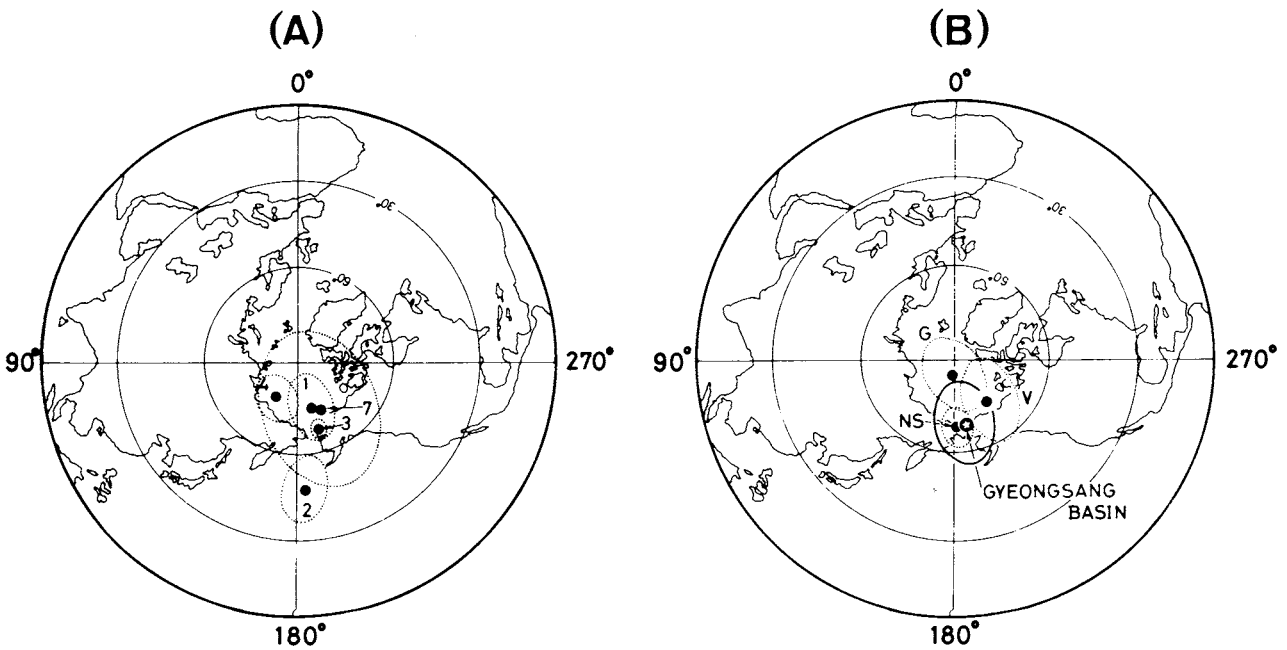


Fig. 3. (*a*) Pole positions and 95% cones of confidence for Gyeongsang Basin. Site numbers are as in Table 2. (*b*) Cretaceous pole positions and 95% cones of confidence for the Korean Peninsula: pole V, volcanic rocks [*Kienzle and Scharon,* 1966]; pole NS, North Korean sediments [*Gurarii et al.,* 1966]; and pole G, granitic rocks [*Ito and Tokieda,* 1980].

sent much the same position of the Cretaceous pole. While the Gyeongsang Basin pole is in excellent agreement with the pole from the sediments of North Korea, the poles from the volcanic and granitic rocks are rather far removed from the Gyeongsang Basin pole. The cause of these deviations may be attributed, in part, to the lack of dip correction for data from the volcanic and granitic samples, because the strata of the Gyeongsang Supergroup which the granites intrude and the volcanics overlie have a southward dip with low angles of about 10°–30°. Thus it seems likely that the paleomagnetic pole (longitude 190°E, latitude 69°N) calculated from data of the Gyeongsang Supergroup and North Korean sediments represents a true Cretaceous pole.

Figure 4 shows the Cretaceous poles for neighboring regions of the Korean Peninsula [*McElhinny,* 1973; *Sasajima,* 1981; *Nishida et al.,* 1980; *Khramov et al.,* 1981]. The Korean pole obtained is in good agreement with poles from China, Kolyma, and the Siberian Platform: The polar distances between the Korean pole and the other poles are less than 13°. However, two circles of 95% confidence for the Korean pole and the southwest Japan pole fail to overlap. The polar distance between them is larger than 23° This separation is explained by the southward drifting accompanied by the clockwise rotation of southwest Japan from the Asian continent [*Hilde and Wageman,* 1973; *Kobayashi and Isezaki,* 1976]. The agreement between the Korean pole and poles from the neighboring regions excepting southwest Japan, on the contrary, suggests that the Korean Peninsula has been a part of the Asian continent at least since the Cretaceous and that the peninsula was not subjected to rotational movement relative to the continent.

Discussion

The Korean Peninsula is flanked by the Sea of Japan, which is a typical marginal basin floored by young oceanic crust. The origin of the Sea of Japan has been explained by back-arc spreading since Cretaceous time, based on the geophysical evidence [*Matsuda and Uyeda,* 1971; *Hilde and Wageman,* 1973; *Uyeda and Miyashiro,* 1974; *Isezaki, 1975*]. The spreading may be due to an upwelling thermal diapir [*Karig,* 1971] or evolution of the convective process in the asthenosphere [*Sleep and Toksöz,* 1971; *Andrews and Sleep,* 1974]. The large-scale extension must have been induced to the edge of the Asian continent just before the spreading. The Chugarian rift valley, which extends from Seoul to Wansan, is possibly the remnant of the extension region in the Korean Peninsula [*O. J. Kim,* 1975; *S. G. Kim,* 1980]. The Korean Peninsula, however, did not drift off from the Asian continent as Japan did, as is concluded in the previous section. This implies that the Korean Peninsula is strong enough to resist the active extension.

The resistance to extension is probably attributed to the stiffness of either the crust or the lithosphere beneath the

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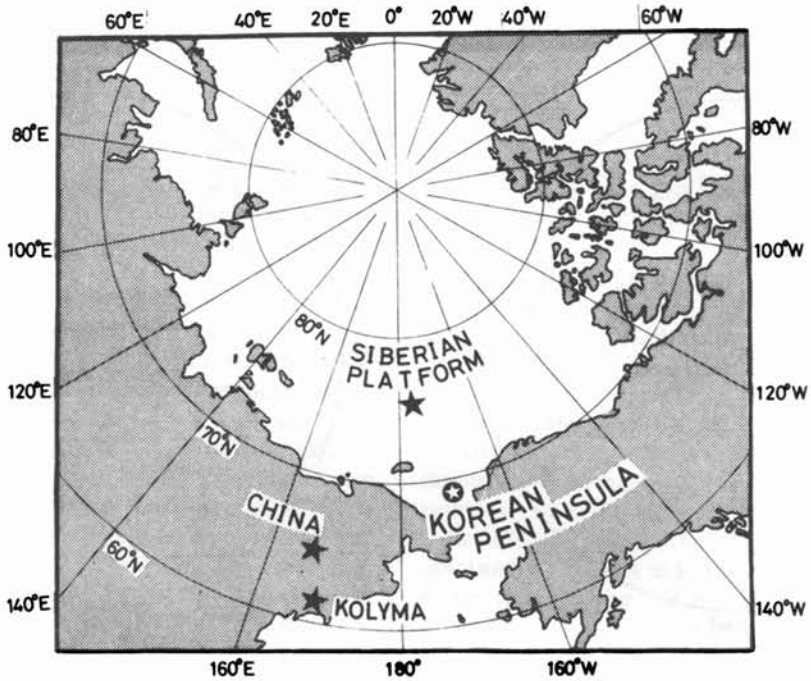


Fig. 4. Cretaceous pole positions for the Korean Peninsula and neighboring regions. The Korean Peninsula pole calculated from data of the present study and North Korean sediments is 189.5°E, 68.9°N. Poles of neighboring regions are a compilation from data of *McElhinny* [1973], *Sasajima* [1981], *Khramov et al.* [1981], and *Nishida et al.* [1981]: Siberian Platform pole (N = 10), 180.9°E, 75.4°N; Kolyma pole (N = 7), 166.1°E, 61.3°N; China pole (N = 8), 164.1°E, 64.0°N; southwest Japan pole (not shown) (N = 11), 196.1°E, 45.9°N. N is the number of poles averaged for the compilation.

Korean Peninsula. The crust beneath the Korean Peninsula is rather thin, less than 30 km [*Cummings and Shiller,* 1971], while the thickness beneath Honshu (Japan) is about 40 km [*Yoshii,* 1979; *Hashizume et al.,* 1981; *Hurukawa,* 1981]. This indicates that the resistance of the landmass against the extension hardly depends on the thickness of the crust.

Honshu has thin lithosphere of about 40 km, and the crust beneath Honshu comes directly into contact with underlying asthenosphere [*Yoshii,* 1979]. On the other hand, the Korean Peninsula has a stiff lithosphere under the crust (lid): The propagation of S_n, a short-period seismic shear wave traveling in the uppermost part of the mantle, is very efficient and shows a much higher Q value than below Honshu [*Molnar and Oliver,* 1969]. Because the Korean Peninsula is part of the Sino-Korean Precambrian shield [*Um and Reedman,* 1975], the thick and stiff lid is also expected from thickening plate theory [*Parker and Oldenburg,* 1973; *Oldenburg,* 1975; *Kono and Amano,* 1978]. This evidence strongly suggests that the stiffness of the lid within the lithosphere controls the occurrence of the isolation of the landmass from the continent. There remains, however, another possible explanation of the stable condition of the Korean Peninsula, that the most active extension area was far from the peninsula.

Conclusion

Five reliable sites from volcanic and sedimentary rocks in the Gyeongsang Basin, in the southeastern part of the Korean Peninsula, have yielded an average Cretaceous pole of 191.2°E, 68.9°N. The representative Korean pole of Cretaceous age, 190°E, 69°N, calculated from Gyeongsang Basin and North Korean sediments is in good agreement with poles from China, Kolyma, and the Siberian Platform excepting southwest Japan. This implies that the Korean Peninsula has been a part of the Asian continent at least since the Cretaceous and that during this time it has not been subjected to any rotational motion relative to the continent. This strong resistance to tectonic disturbances of the Korean Peninsula is probably attributed to the presence of a thick and stiff lid within the lithosphere beneath the peninsula.

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